THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 13 & 14: **Algebra Play, Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** $3n$	$3(n+7)-21=3n$
2	**(b)** $10\times3+7$	Place value
3	**(a)** 88	$11(a+b)$
4	**(d)** 19	Top = a + 2b + c
5	**(b)** $n+12$	$3n+12-2n$
6	**(d)** 120 cm²	$l\times b$
7	**(d)** 70 cm²	$\frac{1}{2}\times b\times h$
8	**(a)** 120 cm²	base × height
9	**(a)** 50 cm²	$\frac{1}{2}(a+b)h$
10	**(a)** 40 cm²	$\frac{1}{2}d_1d_2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	One of two consecutive integers is even (2).	2
12	$100a+10b+c=99a+9b+(a+b+c)$; the first part is divisible by 3 (2).	2
13	35000 cm² (2).	2
14	Side 13; area 169 cm² (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Choose equal or closest numbers (1); 9 and 9 (1); product 81 (1).	1+1+1
16	Area $=\frac{1}{2}\times18\times12=108$ m² (2); cost ₹ 1296 (1).	2+1
17	Choose equal or closest numbers (1); 11 and 11 (1); product 121 (1).	1+1+1
	OR: Choose equal or closest numbers (1); 15 and 15 (1); product 225 (1).	1+1+1
18	Area $=\frac{1}{2}\times25\times14=175$ m² (2); cost ₹ 2100 (1).	2+1
	OR: Area $=\frac{1}{2}\times30\times8=120$ m² (2); cost ₹ 1440 (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $(n-1)(n+1)$ is a product of two consecutive even numbers (3). (ii) $80=8\times10$ (2).	3+2
	OR: (i) Pair opposite entries symmetrically (3). (ii) Sum 135 (2).	3+2
20	(i) 360 cm² (2). (ii) Foot distance 8 cm; area $\frac{1}{2}\times8\times15=60$ cm² (3).	2+3
	OR: (i) 10×10=100, 12×8=96, 15×5=75, 18×2=36 (3). (ii) 10 × 10 square (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 13 & 14: **Algebra Play, Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** equal (or as close as possible)	Largest product
2	**(a)** $n + 2$	(3n+6)/3
3	**(c)** 96	$4x + 16$
4	**(c)** $2n + 1$	Expand
5	**(a)** 5	$(2n + 10)/2 - n = 5$
6	**(b)** 8 cm	Area ÷ length
7	**(b)** perimeter	Perimeter is not a measure of area
8	**(a)** equal area	Same height and equal bases
9	**(b)** 50 cm²	$\frac{1}{2}d^2$
10	**(b)** 6 cm²	Right-angled: ½×3×4

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	4 (2).	2
12	$2n(2n + 2)$ (2).	2
13	9 cm (2).	2
14	42 cm² (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Choose equal or closest numbers (1); 18 and 19 (1); product 342 (1).	1+1+1
16	Area $= \frac{1}{2} \times 22 \times 15 = 165$ m² (2); cost ₹ 1980 (1).	2+1
17	$78 + 87 = 165 = 11 \times 15$ (1.5); $11(a + b)$ (1.5).	1½+1½
	OR: $(10a + b) - (10b + a) = 9(a - b)$ (3).	3
18	Outer $24 \times 20 = 480$ (1.5); path $480 - 320 = 160$ m² (1.5).	1½+1½
	OR: Area 240 cm² (1.5); height $240/13 \approx 18.5$ cm (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $10a + b - (a + b) = 9a$ (2). (ii) $47 - 11 = 36$ (1). (iii) Always a multiple of 9 (2).	2+1+2
	OR: (i) 576 (2). (ii) 1000 (3).	2+3
20	(i) 600 m² (2). (ii) Same area (1). (iii) ₹ 4800 (2).	2+1+2
	OR: (i) Four congruent right triangles (3). (ii) 16 cm (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET C

Chapters 13 & 14: **Algebra Play, Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $5n$	$5(n+4)-20=5n$
2	**(b)** $10\times5+2$	Place value
3	**(b)** 121	$11(a+b)$
4	**(d)** 19	Top = a + 2b + c
5	**(d)** odd	$2n+1$
6	**(a)** 198 cm²	$l\times b$
7	**(c)** 120 cm²	$\frac{1}{2}\times b\times h$
8	**(d)** 63 cm²	base × height
9	**(a)** 160 cm²	$\frac{1}{2}(a+b)h$
10	**(b)** 120 cm²	$\frac{1}{2}d_1d_2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	5, 8; top 13 (2).	2
12	$12\times13=156$ (2).	2
13	5 : 8 (2).	2
14	15 m (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$(n+3)^2-n^2-9=6n$ (2); $6n/n=6$ (1).	2+1
16	Other diagonal 14 cm (1.5); side $\sqrt{12^2+7^2}\approx13.9$ cm (1.5).	1½+1½
17	$2n+(2n+2)+(2n+4)=6n+6=6(n+1)$ (3).	3
	OR: $a+2\times6+c=25$ (1.5); $c=6$ (1.5).	1½+1½
18	Room 48 m², carpet 24 m² (2); uncovered 24 m² (1).	2+1
	OR: Triangle 30 cm² (1); parallelogram 60 cm² (1); ratio 1 : 2 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $(2a+1)+(2b+1)=2(a+b+1)$ (3). (ii) $5+7=12$ (2).	3+2
	OR: (i) $16n$ (3). (ii) 144 (2).	3+2
20	(i) $40+18=58$ units² (3). (ii) Subtract a missing rectangle from a larger one (2).	3+2
	OR: (i) Square 144, rectangle $16\times8=128$ (3). (ii) Square has the larger area (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET D

Chapters 13 & 14: **Algebra Play, Area** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 100	10 × 10
2	**(a)** 36	$4x + 16$
3	**(c)** 140	$4x + 16$
4	**(c)** 1	$n^2 - (n-1)(n+1) = 1$
5	**(d)** $101a + 10b$	100a+10b+a
6	**(d)** 10 cm	Area ÷ length
7	**(b)** square units	e.g. cm²
8	**(b)** equal area	Same base and height
9	**(c)** 1600 m²	40^2
10	**(c)** 24 cm²	Other diagonal 8

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$6n + 9$ (2).	2
12	n^2 (2).	2
13	180 square units (2).	2
14	Other diagonal 24 (1); area 120 cm² (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$20 \times 21 = 420$ (2); nearest equal pair (1).	2+1
16	$196 - 36 = 160$ cm² (3).	3
17	$11(10a + b) = 100a + 10(a + b) + b$ (2); 473 (1).	2+1
	OR: $4x + 16 = 92$ (1.5); $x = 19$ (1.5).	1½+1½
18	$\frac{1}{2}(30)h = 150$ (2); $h = 10$ cm (1).	2+1
	OR: Other leg 24 cm (2); area 84 cm² (1).	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $x + 8$ (1). (ii) Pair symmetric entries (2). (iii) 153 (2).	1+2+2
	OR: (i) $10 \equiv 1 \bmod 9$; $10 \equiv -1 \bmod 11$ (3). (ii) Digits 8,5,7,1,6,2: $(2+1+5) - (6+7+8) = -13$, no (2).	3+2
20	(i) Half difference 20; height $\sqrt{50^2 - 20^2} \approx 45.8$ m (2). (ii) $\frac{1}{2} \times 200 \times 45.8 \approx 4580$ m² (1). (iii) ≈ ₹ 68700 (2).	2+1+2
	OR: (i) Both equal base × common height (3). (ii) Base 10, height 6 gives 60 for both (2).	3+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.